Anomaly detection for data quality, *measured inside* the ETL pipeline

Progress review at the end of M1. Rule-based and classical machine-learning detectors run through one pipeline and are compared per anomaly type, on a held-out split, over five seeds.

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REPO github.com/theobigand/etl-anomaly-benchmark

CODE ~1.7k LOC · 82 tests · CI green

DATASETS Credit Card Fraud, Cardio, Water Quality, synthetic (6 anomaly types)

THE QUESTION

Does machine learning in an ETL pipeline *beat plain rules*, and for which anomalies?

The industry default is rule-based. Declarative checks in tools such as Great Expectations are auditable and cheap, but they scale poorly with schema complexity, only catch violations the author anticipated, and miss multi-column or distributional problems.

Recent work adds ML detectors inside the ETL flow, but the evidence is thin: mostly conceptual architectures with no reproducible code, and when an experiment is reported it is one algorithm on one dataset.

This thesis builds the missing head-to-head benchmark and, as its original contribution, breaks the result down by the kind of anomaly.

Four gaps in generic anomaly benchmarks

EVALUATED IN A VACUUM
Algorithms scored outside any pipeline, no integration cost.

NO DECLARATIVE BASELINE
The rules that dominate production are never in the comparison.

ONE AGGREGATE SCORE
Per dataset, hiding which family is good at what.

NO OPERATING POINT
A single flag threshold, not the precision-recall trade-off.

THE ORIGINAL CONTRIBUTION

A controlled dataset that *labels six anomaly types* separately

`missing value`	one or two feature cells set to null
`exact duplicate`	a clean row copied, same business attributes
`univariate outlier`	one numeric field pushed far out of range
`multivariate outlier`	every field in range, the combination not (an 18-year-old with a top-percentile balance)
`temporal drift`	recent-signup rows whose income and balance distribution has shifted up
`invalid format`	malformed e-mail, out-of-domain country code, impossible age

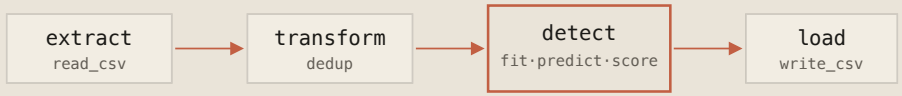
12,360 records generated with Faker and NumPy, six types injected at 3% each, every row tagged with its category. Recall **and** precision are reported per type.

The injection magnitudes are exposed as one **severity** multiplier, so the choice of magnitude is swept rather than fixed.

A time-aware `drift` detector is added for the drift column; it runs on the synthetic dataset only.

THE SYSTEM

One interface, *one measured path*



The demo and every benchmark call the same `detect()` stage, so latency is the real in-pipeline fit + predict + score time.

A continuous `score()` on top of the boolean `predict()` makes ROC-AUC and PR-AUC computable. `detect(fit_df=...)` fits on the train slice and scores the test set.

```python
# every detector, one contract
class Detector(ABC):
    name: str
    def fit(self, df) -> Self: ...
    def predict(self, df) -> Series:   # bool per row
        ...
    def score(self, df) -> Series:     # graded score
        return self.predict(df).astype(float)

# classical ML: adapt PyOD, do not reimplement
class PyODDetector(Detector):
    def fit(self, df):
        x = impute(df[self.numeric_cols])
        self.model = self.estimator.fit(scale(x))
        return self

# registry: one line, picked up everywhere
"isolation_forest": lambda: IsolationForestDetector(),
```

src/etl_benchmark/pipeline.py
quality/base.py

python -m etl_benchmark --detector <name>

interface modelled on PyOD BaseDetector

TWO FAMILIES NOW, A THIRD PLANNED

Seven detectors, *one contract*

Detector	Family	Principle	Continuous score	Main limit
Z-score	statistical rule	flag `|z| > 3` on any column	worst `|z|` across columns	mean and std inflated by outliers
IQR	statistical rule	Tukey fences, `k = 1.5`	largest fence exceedance	any-column rule fires often in high dim
Great Expectations	declarative rule	not-null + range at 0.1 / 99.9 pct	fraction of the suite violated	only checks what the suite declares
Isolation Forest	classical ML	isolation depth under random splits	PyOD `decision_function`	axis-aligned splits, weak on nulls
Local Outlier Factor	classical ML	local density vs 20 neighbours	PyOD `decision_function`	scoring cost grows with the scored set
One-Class SVM	classical ML	RBF boundary around the bulk	PyOD `decision_function`	quadratic training
drift	time-aware	recent window mean shift, per-row lean	`w · lean`	per-row, so diluted drift is invisible

Classical ML detectors are wrapped from PyOD. Flag threshold from the train-set prevalence. Deep learning (Autoencoder, LSTM-AE) is M2.

FOUR DATASETS, ONE PROTOCOL

Three real settings *bracket the method*

Dataset	Rows	Feat.	Anom.	Role
Credit Card Fraud	283,726	30	0.17%	rare, but a mixed-defect label
Cardio	1,831	21	9.6%	rare and outlying, the fitting regime
Water Quality	3,276	9	61%	target class is common, wrong regime
Synthetic per-type	12,360	9	17.5%	controlled, 6 labelled types

Protocol. Stratified 70 / 15 / 15 split, fit on train, every number reported on the held-out test set. Five seeds, mean ± std.

The ML contamination is read from the train-set prevalence, not fixed at 10%. That one change moves Isolation Forest's fraud precision from 0.01 to 0.20.

LOF and One-Class SVM run everywhere except Credit Card Fraud, where neither scales to 284k rows in the pipeline.

HOW EACH ANOMALY TYPE IS INJECTED

Magnitudes are a *swept parameter*, not a fixed choice

`missing value`	1–2 of {income, balance, avg amount, tenure, n txn, email, country} → null
`duplicate`	exact copy of a sampled clean row, id included
`univariate`	one numeric column × U(15, 45) × severity
`multivariate`	marginals at the (1 − 0.03/severity) percentile of the clean data
`temporal drift`	signup in the last 80 days; income × (1 + 0.55·severity), similar for balance and amount
`invalid format`	email → malformed; age → impossible; country → out of domain

severity = 1.0 is the reference dataset. Missing values, duplicates and invalid formats have no magnitude to scale.

The severity sweep (slide 15) turns "we picked the magnitudes" into a curve and shows the per-type ranking is stable.

Five types replace clean rows; duplicates are appended.

THE PIVOT

Aggregate metrics say the detector families are interchangeable.
Per anomaly type, they are not.

F1 spans 0.31 to 0.51 on the synthetic set · the per-type competences barely overlap

REAL FRAUD DATA

The *operating point* decides the outcome

Detector	Prec	Rec	F1	ROC	PR	Flag%	Time
Z-score	0.011	0.870	0.021	0.926	0.078	13.4	**0.03s**
IQR	0.003	**0.966**	0.007	0.911	0.018	48.6	0.12s
Great Expectations	0.037	0.676	0.070	0.830	**0.182**	3.1	1.37s
Isolation Forest	**0.202**	0.206	**0.203**	**0.934**	0.120	**0.2**	1.78s

Mean over 5 seeds, 15% held-out test split.

Isolation Forest, by contamination	Prec	Rec	Flag%
fixed 0.10 (before)	0.014	**0.88**	10.4
= train prevalence (now)	**0.202**	0.21	0.2

The statistical rules recover most frauds at precision below 0.04: a fixed per-column rule cannot separate 0.17% positives from the tails of normal traffic.

Isolation Forest changes once its contamination matches the prevalence: precision **0.20**, the first usable operating point, at recall 0.21 and with real seed variance (F1 = 0.20 ± 0.07).

Great Expectations leads PR-AUC (0.182 ± 0.046). LOF and One-Class SVM do not scale to 284k rows here.

GENUINELY IMBALANCED REAL DATA

When anomalies are *rare and outlying*, the learned detectors work

Detector	Prec	Rec	F1	ROC	PR	Flag%
Z-score	0.49	**0.82**	0.61	0.93	0.61	16.6
IQR	0.21	0.90	0.34	0.78	0.30	43.4
Great Expectations	0.51	0.21	0.29	0.59	0.19	4.6
Isolation Forest	**0.64**	0.62	**0.62**	**0.95**	**0.70**	9.9
LOF	0.25	0.27	0.26	0.62	0.24	10.7
One-Class SVM	0.56	0.61	0.58	0.94	0.63	10.7

Cardiotocography, 1,831 rows, 9.6% anomalies. Mean over 5 seeds, test split.

Isolation Forest reaches **F1 0.62** and **PR-AUC 0.70**. Z-score also does well: the 21 features are standardized and anomalies stand out on several at once.

This is the counterpoint to the other two real datasets: value shows up when the anomalies are genuinely rare outliers, not a 0.17% mixed label or a 61% majority.

Wide seed spread on this small set (ML F1 std up to 0.17).

experiments/run_benchmark.py
--dataset cardio

make benchmark-cardio

the regime the detectors are built for

SYNTHETIC DATASET, AGGREGATE

Aggregate metrics make the families *look interchangeable*

Detector	Prec	Rec	F1	ROC	PR	Time
RULE-BASED						
Z-score	**1.000**	0.182	0.308	**0.748**	**0.560**	**0.002s**
IQR	0.603	0.399	0.481	0.678	0.460	0.004s
Great Expectations	0.920	0.205	0.335	0.601	0.332	0.29s
CLASSICAL ML						
Isolation Forest	0.515	**0.503**	**0.509**	0.745	0.501	0.16s
Local Outlier Factor	0.380	0.390	0.385	0.646	0.437	0.14s
One-Class SVM	0.488	0.478	0.483	0.729	0.531	2.61s

Mean over 5 seeds, test split. 12,360 rows, 17.5% anomalous.

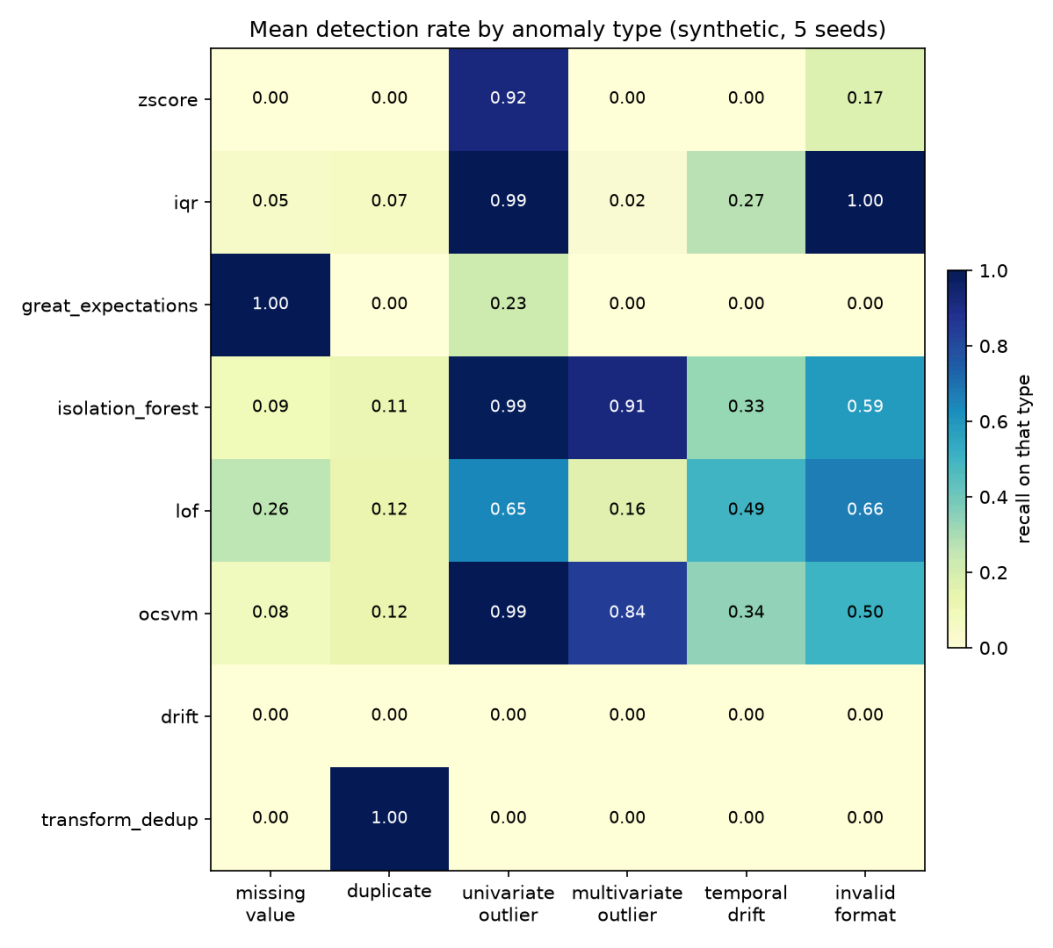
F1 spans **0.31 to 0.51**, PR-AUC **0.33 to 0.56**. Isolation Forest has the best F1; no family is clearly ahead.

Z-score keeps a perfect precision by firing only on the most extreme rows. One-Class SVM matches Isolation Forest at about **16×** the latency.

The per-type view on the next two slides shows the opposite.

SYNTHETIC DATASET, PER TYPE — THE CONTRIBUTION

Each method has a *narrow competence*



WHAT IT SHOWS

Mean recall over 5 seeds of each method on each injected type. Darker is higher.

READING IT

Z-score owns univariate outliers. Great Expectations owns missing values (recall 1.00, precision 0.90). Deduplication owns exact duplicates. Isolation Forest (0.91) and One-Class SVM (0.84) are the only methods that catch multivariate outliers, at about 0.24 precision. Temporal drift is the weakest column for everyone.

WHY IT MATTERS

The competences barely overlap. Broad coverage needs several families, not the single best one on aggregate.

paper/figures/
heatmap_synthetic.png

transform_dedup and drift shown for reference

central figure of the paper

RECALL AND PRECISION, PER TYPE

Detection rate alone *can be gamed*

recall (detection rate)

Method	miss	dup	uni	mul	drift	fmt
Z-score	0.00	0.00	**0.92**	0.00	0.00	0.17
IQR	0.05	0.07	**0.99**	0.02	0.27	**1.00**
Great Exp.	**1.00**	0.00	0.23	0.00	0.00	0.00
Isolation Forest	0.09	0.11	**0.99**	**0.91**	0.33	0.59
LOF	0.26	0.12	0.65	0.16	**0.49**	0.66
One-Class SVM	0.08	0.12	**0.99**	0.84	0.34	0.50
drift	0.00	0.00	0.00	0.00	0.00	0.00
transform dedup	0.00	**1.00**	0.00	0.00	0.00	0.00

precision (clean rows as the only negatives)

Method	miss	dup	uni	mul	drift	fmt
Z-score	–	–	**1.00**	–	–	**1.00**
IQR	0.03	0.04	0.39	0.01	0.15	0.39
Great Exp.	**0.90**	–	**0.70**	–	–	0.05
Isolation Forest	0.03	0.04	0.26	0.24	0.10	0.17
LOF	0.06	0.03	0.15	0.04	0.11	0.15
One-Class SVM	0.03	0.04	0.25	0.22	0.10	0.14
drift	–	–	–	–	–	–
transform dedup	–	0.52	–	–	–	–

Isolation Forest catches 91% of multivariate outliers at only 0.24 precision: a wide net that also pulls in clean rows. Z-score fires only when it is certain, so its precision is 1.00 on the two columns it covers.

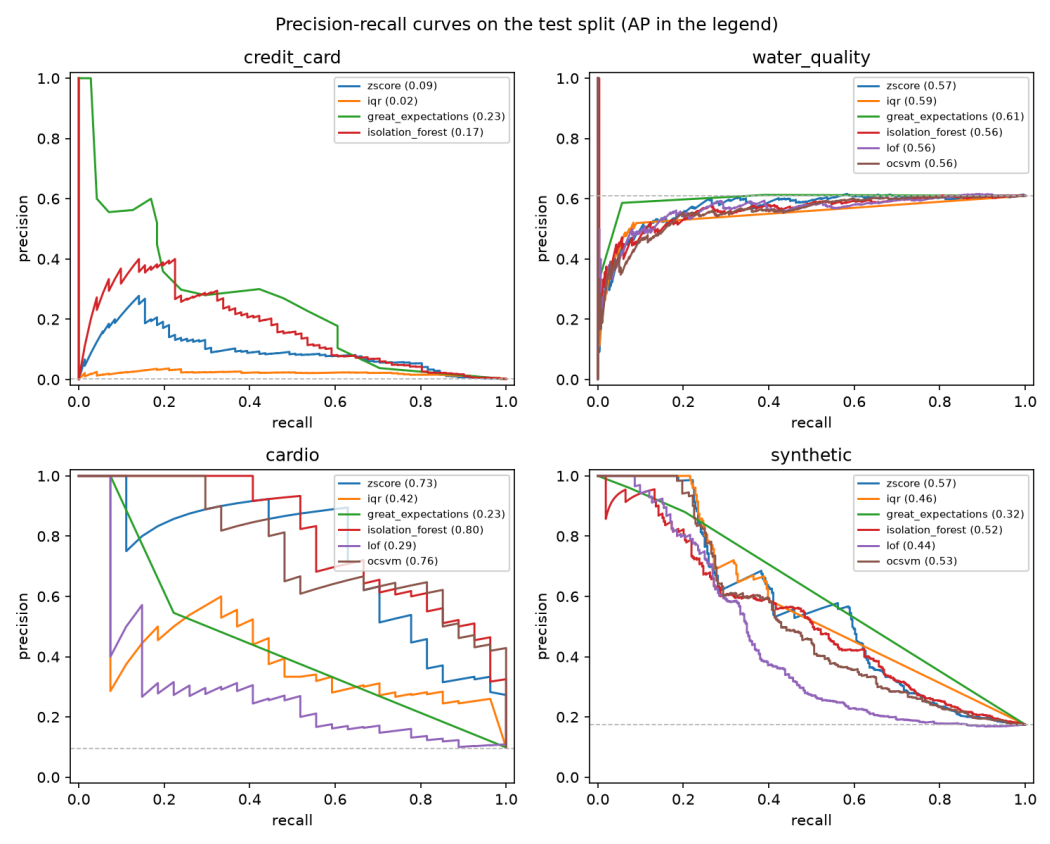
experiments/results/
synthetic_recall / _precisi
on

synthetic_recall / synthetic_precision CSVs

5 seeds, mean

PRECISION–RECALL CURVES AND THE OPERATING POINT

One threshold hides *the whole trade-off*



READING IT

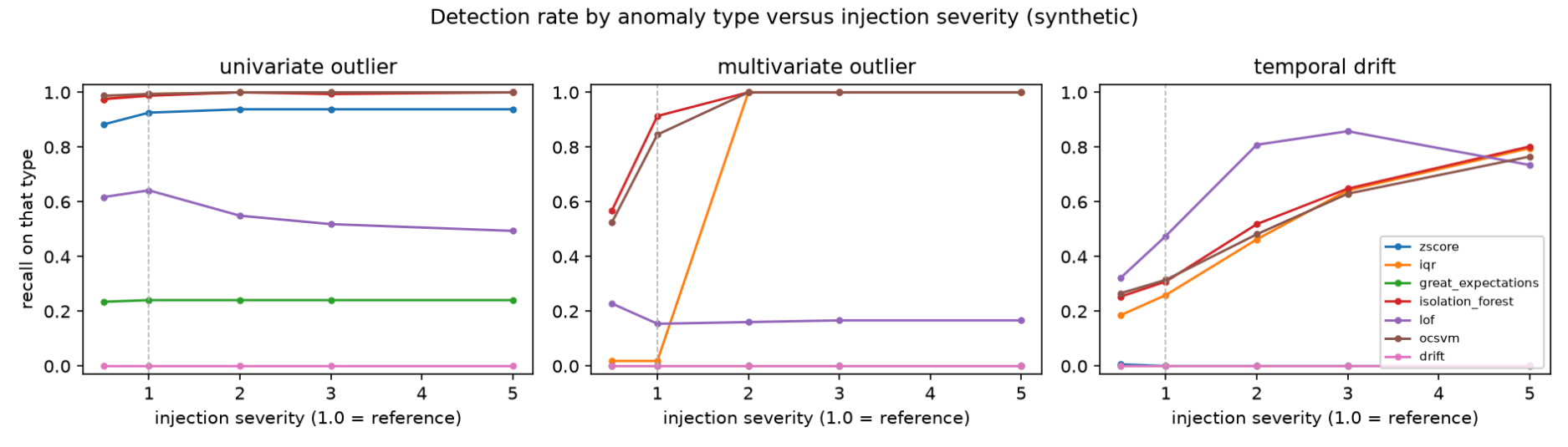
On Cardio the Isolation Forest and One-Class SVM curves stay high (AP 0.80, 0.76). On Water Quality every curve hugs the 0.61 base-rate line. On Credit Card Fraud Great Expectations and Isolation Forest dominate but neither clears precision 0.4 except at negligible recall.

CONTAMINATION SWEEP, ISOLATION FOREST

Fraud: precision 0.37 → 0.008 as recall goes 0.23 → 0.92. Cardio: precision 1.00 at recall 0.04, or recall 0.93 at precision 0.46. Same estimator, only the threshold moves. Reading the prevalence lands near the knee where one exists.

DETECTION RATE VERSUS INJECTION SEVERITY

The per-type ranking is *stable* as the magnitudes scale



Univariate outliers are saturated already at severity 0.5. Multivariate outliers show a threshold: Isolation Forest and One-Class SVM climb to 1.0; IQR jumps once the injected percentile crosses its fence.

Temporal drift climbs roughly linearly for the distributional detectors, from ≤ 0.3 at severity 1.0 to 0.6–0.85 at severity 3. The reference drift is simply below every method's floor, not a property of the methods. The same families lead the same columns at every severity.

`experiments/run_severity.py`
`paper/figures/severity_curves.png`

`make severity`

`answers the "we picked the magnitudes" threat`

THE PRACTICAL OUTPUT

Anomaly type *to method*

Anomaly type	Use	Recall	Why
missing value	Great Expectations	1.00	not-null expectation, exact
exact duplicate	transform dedup stage	1.00	row equality, not a detector's job
univariate outlier	any statistical rule	0.92–0.99	single-column band is enough
multivariate outlier	Isolation Forest	0.91	only the joint models see it (precision 0.24)
temporal drift	windowed monitor	≤0.49	sub-threshold per row; needs a stage, not a detector
invalid format	IQR, partially	1.00 / none	numeric part caught, string format missed

Two of six types are not a per-row detector problem at all: duplicates belong to the transform stage, drift to a windowed monitor. That itself is a finding a single aggregate ranking would never surface.

OPERATIONAL COST

Latency and scaling *constrain the shortlist*

Method	Fraud test	Synth. test	Scaling
Z-score	0.03s	0.002s	linear
IQR	0.12s	0.004s	linear
Great Expectations	1.37s	0.29s	linear, one pass per expectation
Isolation Forest	1.78s	0.16s	linear
Local Outlier Factor	n/a	0.14s	superlinear in the scored set
One-Class SVM	n/a	2.61s	quadratic training

Detect time on the 15% test split. GX takes ~13s on the full fraud dataset.

A detector has to sit in the flow, so wall-clock time is part of the comparison.

Great Expectations is the slowest linear method. **One-Class SVM** and **LOF** do not survive contact with the full fraud dataset in the pipeline.

A benchmark scoring algorithms in isolation would not surface this.

WHAT WAS HARDER THAN THE PLAN ASSUMED

Findings *from the build*

THE OPERATING POINT MATTERS AS MUCH AS THE ALGORITHM

Setting Isolation Forest's contamination to the data prevalence instead of a fixed 10% moves its fraud precision from 0.01 to 0.20 and its multivariate-outlier recall from 0.41 to 0.91. The same estimator, one number changed.

UNSUPERVISED DETECTION WORKS IN ONE REGIME

Cardio (rare, outlying): Isolation Forest F1 0.62. Credit Card Fraud (0.17% mixed label): nothing usable. Water Quality (61% majority): everyone at chance. The question is less "rules or ML" than "does this dataset have rare outliers".

GREAT EXPECTATIONS 1.X IS A FULL API REWRITE

The planning notes assumed the 0.18 API. Getting per-row failures out of GX 1.18 needs the COMPLETE result format plus an explicit index column. Figured out by probing the installed library.

ONE-CLASS SVM DOES NOT SCALE TO PIPELINE DATA

Quadratic training, and LOF scoring is superlinear in the scored set. On the full fraud dataset only rules and Isolation Forest run.

ML DETECTORS ARE WEAK ON MISSING VALUES

Imputing before the model removes the signal a not-null rule catches for free: GX recall 1.00 on that type against 0.08 to 0.26 for the learned models.

DRIFT IS NOT A PER-ROW PROBLEM

Once diluted by clean recent rows the window mean has moved by about a tenth of a standard deviation. The dedicated detector scores 0 at the reference severity; the severity sweep shows it needs a several-times-larger shift.

IMPLEMENTED AND TESTED

What the repository *holds today*

Module	Tests	Covers
quality/rules, gx	13	flags, constant col, score, null + range
quality/ml	16	3 detectors: planted outliers, ranking, imputation
quality/drift	5	window shift, no-time no-op, index
split, registry, datasets	16	ratios, class balance, contamination flow, specs
synthetic	12	reproducible, all types, severity monotone
run_benchmark, run_synthetic_benchmark	8	mean ± std over seeds, per-type precision, figure
pipeline	4	label excluded, fit-sample, timings

82 tests, ruff clean, CI on push. conda env, Makefile, versioned paper PDF.

Since the last review:

FEAT(PIPELINE)
stratified split, prevalence contamination

FEAT(DATA)
Cardio, Water Quality, severity knob

FEAT(QUALITY)
time-aware drift detector

FEAT(EXPERIMENTS)
multi-seed, per-type precision, PR + severity curves

CI
GitHub Actions: ruff + pytest

WHERE IT STANDS

Rigour gaps closed, *substance left for M2*

Item	Status	Detail
held-out split, 5 seeds, mean ± std	DONE	metrics on the test split only
contamination from prevalence	DONE	plus a full contamination sweep
per-type precision	DONE	alongside recall, stable under a severity sweep
imbalanced real dataset	DONE	Cardio; more would firm up the generalisation claim
CI	DONE	GitHub Actions, ruff + pytest
deep learning	M2	Autoencoder, then LSTM-Autoencoder, same interface
temporal drift	OPEN	a windowed monitor, not the per-row detector used here
Related Work section	M2	plus bibliography checks, then the final Discussion

Every rigour gap from the last review is closed. Deep learning and the write-up are the work left for M2. Defense July 2027.

TAKEAWAY

Rules and machine learning are *complementary by anomaly type.*

The per-type heatmap is what makes that choice actionable. Across three real datasets, unsupervised detection pays off only on Cardio, where the anomalies are genuinely rare outliers. The pipeline, the four datasets, the held-out protocol, the curves and the CI are in place and tested. Deep learning and the write-up are the work for M2.